free of any baggage inherited from our classical experience, is extraordinarily helpful when tackling the deep problems of modern physics.

Given the necessity of these two ingredients (wave functions and the Schrödinger equation), there are a few alternatives to Many-Worlds we might also consider. One is to imagine adding new physical entities over and above the wave function. This approach leads to hidden-variable models, which were in the back of the minds of people like Einstein from the start. These days the most popular such approach is called the *de Broglie–Bohm theory*, or simply *Bohmian mechanics*. Alternatively, we could leave the wave function by itself but imagine changing the Schrödinger equation, for example, to introduce real, random collapses. Finally, we might imagine that the wave function isn't a physical thing at all, but simply a way of characterizing what we know about reality. Such approaches are broadly known as epistemic models, and a currently popular version is *QBism*, or *quantum Bayesianism*.

All of these options—and there are many more not listed here—represent truly distinct physical theories, not simply "interpretations" of the same underlying idea. The existence of multiple incompatible theories that all lead (at least thus far) to the observable predictions of quantum mechanics creates a conundrum for anyone who wants to talk about what quantum theory really means. While the quantum recipe is agreed upon by working scientists and philosophers, the underlying reality—what any particular phenomenon actually *means*—is not.

I am defending one particular view of that reality, the Many-Worlds version of quantum mechanics, and for most of this book I will simply be explaining things in Many-Worlds terms. This shouldn't be taken to imply that the Everettian view is unquestionably right. I hope to explain what the theory says, and why it's reasonable to assign a high credence to it being the best view of reality we have; what you personally end up believing is up to you.

---

* If you look on the Internet, you will find numerous attributions of "Shut up and calculate!" to Richard Feynman, a physicist who was an all-time great at doing difficult calculations. But he never said any such thing, nor would he have found the sentiment congenial; Feynman thought carefully about quantum mechanics, and nobody ever accused him of shutting up. It's common for quotations to be reattributed to plausible speakers who are more famous than the actual source of the quote. Sociologist Robert Merton has dubbed this the Matthew Effect, after a line from the Gospel of